

T3 matched EFT Lagrangian terms: EFT_2_after_S1

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1. Remaining active T3 fields: S2.

Field combinations 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	$S_2 D$	$H \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-S_{2,j_1} S_{2,j_1}^\dagger M_{S_2}^2$ $+\frac{2h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_2}^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger\right)^2}{3\epsilon}$ $-\frac{C_{S_2^\dagger S_2} h\overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{2C_{S_2^\dagger S_2} \frac{\sqrt{3}}{\sqrt{3}\epsilon}}$	$+D_{\mu_1} \left(S_{2,j_1}\right)^\dagger D_{\mu_1} \left(S_{2,j_1}\right)$	$+\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$
cont.								$-\frac{4h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1 \lambda_{S_2}^{(1)}$ $-\frac{2h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1 \lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$		$+\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$
cont.								$-\frac{h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1 \lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2 \lambda_{S_2}^{(2)}$ $-\frac{2h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2 \lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$		
cont.								$-\frac{h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2 \lambda_{S_2}^{(2)}}{\sqrt{3}}$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-S_{2,j_1} S_{2,j_1}^\dagger M_{S_2}^2$ $-\frac{1}{2} C_{S_2^\dagger S_2} h\overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2} h\overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$	$+D_{\mu_1} \left(S_{2,j_1}\right)^\dagger D_{\mu_1} \left(S_{2,j_1}\right)$	$+\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$
cont.								$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1 \lambda_{S_2}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1 \lambda_{S_2}^{(1)}}{\epsilon}$		$+\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$
cont.								$-\frac{1}{2} h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1 \lambda_{S_2}^{(1)}$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-S_{2,j_1} S_{2,j_1}^\dagger M_{S_2}^2$ $-\frac{1}{2} \sqrt{3} C_{S_2^\dagger S_2} h\overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$ $-\frac{\sqrt{3} C_{S_2^\dagger S_2} h\overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{3\epsilon}$ $-\frac{5h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$	$+D_{\mu_1} \left(S_{2,j_1}\right)^\dagger D_{\mu_1} \left(S_{2,j_1}\right)$	$-\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3\epsilon M_F}$ $-\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$
cont.								$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1 \lambda_{S_2}^{(1)}$ $-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1 \lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$		$-\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$
cont.								$-\frac{1}{2} \sqrt{3} h\overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1 \lambda_{S_2}^{(1)}$		
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-S_2 S_2^\dagger M_{S_2}^2$ $-\frac{2h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger\right)^2}{\epsilon}$ $-C_{S_2^\dagger S_2} h\overline{y_{2,i_1,r_1}} S_2 y_{2,i_1,r_1} S_2^\dagger$	$+D_{\mu_1} \left(S_2\right)^\dagger D_{\mu_1} \left(S_2\right)$	$+\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$

[illegible]

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	H	ℓ	e
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+\frac{2i\sqrt{\frac{2}{3}}\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^{(1)}}{\epsilon}$	$+\frac{2}{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}$ $-\frac{2C_{S_1^1S_2}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}}{\sqrt{3}}$ $-\frac{4}{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}$	$+\frac{\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}$ $+\frac{3\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\overline{3hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}$ $+\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
cont.								$+2i\sqrt{\frac{2}{3}}\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^{(1)}$ $+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^{(1)}2\lambda_{H2}^{(2)}}{2\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^{(1)}2\lambda_{H2}^{(2)}}}$	$-\frac{2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_3,j_1,j_4,j_2}^{(1)}\lambda_{S_2}^{(1)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_3,j_1,j_4,j_2}^{(2)}\lambda_{S_2}^{(2)}}}{\sqrt{3}}$	$+\frac{3\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$ $+\frac{3}{8}\frac{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$		
cont.								$-\frac{\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^{(2)}\lambda_{H2}^{(2)}}}{\sqrt{3}}$				$+\frac{3}{8}\frac{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}I_{y_{2,a_1,j_1}}^*y_{2,a_2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$ $-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}I_{y_{2,a_1,j_1}}^*y_{2,a_2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$ $+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}{2\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}}$	$-C_{S_2^1S_2}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}}{\overline{y_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}}$ $-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}$ $-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_1,j_3,j_4,j_2}^{(1)}\lambda_{S_2}^{(1)}}}{\overline{y_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_1,j_3,j_4,j_2}^{(1)}\lambda_{S_2}^{(1)}}}$	$+\frac{\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}$ $+\frac{3}{8}\frac{\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}$ $+\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
cont.								$-\frac{1}{2}\frac{\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}}{\epsilon}$ $+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}2\lambda_{H2}^{(2)}}{2\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}2\lambda_{H2}^{(2)}}}$	$-\frac{1}{2}\frac{\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$ $-\frac{1}{2}\frac{\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$	$+\frac{3}{8}\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$ $+\frac{3}{8}\frac{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$		
cont.								$-\frac{1}{2}\frac{\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$ $-\frac{1}{2}\frac{\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$				$+\frac{3}{8}\frac{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{(1)}}{\sqrt{3}\epsilon}$	$-\sqrt{3}C_{S_1^1S_2}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}}{\overline{y_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}}$ $-\frac{5}{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}$ $-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_1,j_3,j_4,j_2}^{(1)}\lambda_{S_2}^{(1)}}}{\overline{y_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_1,j_3,j_4,j_2}^{(1)}\lambda_{S_2}^{(1)}}}$	$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}$ $+\frac{3\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
cont.								$-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{(1)}}{\sqrt{3}}$ $+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}{2\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}}$ $-\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}}{\epsilon}$	$-\frac{1}{2}\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}\lambda_{H2}^{(1)}}}{\epsilon}$ $+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}2\lambda_{H2}^{(2)}}{2\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)}2\lambda_{H2}^{(2)}}}$ $-\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$	$+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$ $+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$		
cont.								$-\frac{1}{2}\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$ $-\frac{1}{2}\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)}\lambda_{H2}^{(2)}}}{\epsilon}$				$+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{\overline{3hy_{2,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,y_2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_2^\dagger}{\epsilon}$ $-\frac{2\overline{hy_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,y_2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_2^\dagger}{\epsilon}$ $+\frac{H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger\mathcal{I}_{H2,a_1,a_2}^{(1)}\lambda_{H2}^{(1)}}{2\overline{hy_{2,i_1,r_1}H_{a_2}S_2y_{2,i_2,r_1}H_{a_1}^\dagger S_2^\dagger\mathcal{I}_{H2,a_1,a_2}^{(1)}\lambda_{H2}^{(1)}}}$	$-2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,y_2,i_2,r_1}^2y_{2,i_2,i_2,r_1}\left(S_2^\dagger\right)^2}{\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,y_2,i_2,r_1}^2y_{2,i_2,i_2,r_1}\left(S_2^\dagger\right)^2}$ $-2C_{S_1^1S_2}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}S_2y_{2,i_1,r_1}S_2^\dagger}}{\overline{y_{2,i_1,r_1}S_2y_{2,i_1,r_1}S_2^\dagger}}$ $-2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}S_2^2y_{2,i_1,r_1}\left(S_2^\dagger\right)^2\lambda_{S_2}^{(1)}}}{\overline{y_{2,i_1,r_1}S_2^2y_{2,i_1,r_1}\left(S_2^\dagger\right)^2\lambda_{S_2}^{(1)}}}$	$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}$ $+\frac{\overline{hy_{2,i_3,r_1}L_{a_1,i_2}}PR e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}}\epsilon_{i_2}PL L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		

[illegible]

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_2	F	H	ℓ	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{+\overline{\lambda T_3} S_{2,j_1} \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger} - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	
								$- \frac{\frac{1}{2} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger} - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,a_1,a_2,j_2,j_1}^{(2)} \lambda_{H2}^{(2)}}{\sqrt{3}}$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_3} L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	
								$- \frac{\sqrt{3} \epsilon}{2\sqrt{3}}$							
								$\frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{2\sqrt{3}}$							
cont.								$+ \frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{2} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3} \epsilon}$							
								$- \frac{\sqrt{3} \epsilon}{2\sqrt{3}}$							
								$\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\sqrt{3}}$							
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$- \frac{+\overline{\lambda T_3} S_{2,j_2} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I} y_{2,j_2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	
								$- \frac{1}{4} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_3} L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	
								$- \frac{1}{4} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$							
								$- \frac{1}{4} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$							
cont.								$+ \frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$- \frac{+\overline{\lambda T_3} S_{2,j_2} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$						$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3 M_F}$	
								$- \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_3} L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3 M_F}$	
								$- \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$							
								$- \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3a_1,a_2,j_1,j_2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$							
cont.								$+ \frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{\sqrt{3} h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{\sqrt{3} h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
								$- \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger}{\overline{\sqrt{3} h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger} - \frac{1}{4} \overline{\sqrt{3} h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$							
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$- \frac{+\overline{\lambda T_3} S_2 \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger} - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\sqrt{3}}$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	
								$- \frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{2\sqrt{3}}$						$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_3} L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	
								$- \frac{\sqrt{3} \epsilon}{\epsilon}$							
								$\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\epsilon}$							
cont.								$- \frac{1}{2} \overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger$							
								$+ \overline{\lambda T_3} S_2 \mathcal{I} T_{3a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$							

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_2	F	H	F	H	ℓ
cont.										$\frac{h\overline{y}_{1,i_1,r_1}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$						
										$\frac{h\overline{y}_{1,i_1,r_1}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$						
										$\frac{h\overline{y}_{2,i_1,r_1}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\epsilon}$						
cont.										$-\frac{1}{2}h\overline{y}_{2,i_1,r_1}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$						
T3-E	0	3	0	3	1	2	$\frac{1}{2}$			$\frac{+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{h\overline{y}_{1,i_1,r_1}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}$	$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$					
										$\frac{h\overline{y}_{1,i_1,r_1}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$+2i\sqrt{\frac{2}{5}}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}{}^\dagger$					
										$\frac{h\overline{y}_{1,i_1,r_1}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$	$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}H_{2,a_1,a_2,j_2,j_1}{}^{(2)}\lambda_{H2}^{(2)}}{\sqrt{3}}$					
cont.										$\frac{h\overline{y}_{2,i_1,r_1}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$					$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}{}^T\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	
										$\frac{h\overline{y}_{2,i_1,r_1}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$					$-2\sqrt{\frac{2}{3}}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{\lambda_{T3}}L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	
										$\frac{h\overline{y}_{2,i_1,r_1}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$					M_F	
cont.										$\frac{+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{h\overline{y}_{1,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}$						
										$\frac{h\overline{y}_{1,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$						
										$\frac{h\overline{y}_{1,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\sqrt{3}}$						
cont.										$\frac{h\overline{y}_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$						
										$\frac{h\overline{y}_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$						
										$\frac{h\overline{y}_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\sqrt{3}}$						

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F H$	$F H \ell e$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}$
cont.								$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}$
cont.								$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{1}{2} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{1}{2} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{1}{2} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}$
cont.								$-\frac{1}{2} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}$
cont.								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}$	$+\frac{1}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}$
								$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}$
cont.								$-\frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}}{\sqrt{3}}$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}$